

Tide retrospective: *the connected cumulants in moment form*

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2026-05-31

Abstract

A polish tide closing the presentation of the anharmonic cumulant arc: the third and fourth connected cumulants of $u = -(tx)$, proved earlier as rational forms in the partition values $G_n(0)$, re-expressed in the physically transparent unperturbed-Gibbs moment form,

$$\kappa_3 = \langle u^3 \rangle_0 - 3\langle u^2 \rangle_0 \langle u \rangle_0 + 2\langle u \rangle_0^3,$$

$$\kappa_4 = \langle u^4 \rangle_0 - 4\langle u^3 \rangle_0 \langle u \rangle_0 - 3\langle u^2 \rangle_0^2 + 12\langle u^2 \rangle_0 \langle u \rangle_0^2 - 6\langle u \rangle_0^4,$$

with $\langle u^k \rangle_0 = \text{gibbsExp} \cdots (\cdot^k)$. New file `AnharmonicCumulantMomentForms.lean`; builds clean, no `sorry/axiom/native_decide`.

1 Setting

The anharmonic cumulant arc proved $\kappa_2, \kappa_3, \kappa_4$ as derivatives of the perturbed mean and identified them with $\partial^n \log Z$. The cumulant *values*, however, were stated as rational expressions in the bare partition integrals $G_n(0)$ — correct but opaque. This tide re-states κ_3, κ_4 in the form a reader of the Susceptibility Primer expects: combinations of the raw moments $\langle u^k \rangle_0$ under the unperturbed Gibbs measure. (κ_2 's moment form, `anharmonic_susceptibility_eq_connected_variance`, already existed.) It was the lowest-risk available step: warm seabed, zero race, a definitional re-expression of already-proved theorems.

2 The thing formalised

For the anharmonic potential, κ_3 and κ_4 (as the second and third iterated derivatives of the mean G_1/G_0 at 0) equal the moments-to-cumulants combinations displayed above, with $\langle u^k \rangle_0 = \text{gibbsExp volume } L \text{ id } t \ 0 \ (x \mapsto -(tx))^k$.

3 Proof strategy

A single helper does the work: moment-normalisation gives $\langle u^k \rangle_0 = G_k(0)/G_0(0)$ for every k (compose `iteratedDeriv_partition_div_eq_gibbsExp` with the `iteratedDeriv` identity). Rewriting every $\langle u^k \rangle_0$ on the right-hand side to $G_k(0)/G_0(0)$, and the cumulant on the left to its proved $G_n(0)$ -rational form, leaves a pure scalar identity over the common denominator $G_0(0)$, which `field.simp` (clearing $G_0(0) \neq 0$) closes.

4 Roadblocks and resolutions

Only one, trivial: after `field_simp` the goal was already closed, so the `ring` I had appended reported “no goals”. `field_simp` fully normalises and discharges these moment identities on its own; the `ring` was simply removed. (Contrast the κ_4 derivative tide, where `field_simp` did *not* close and `ring` was load-bearing — the behaviour is identity-dependent.)

5 What was Mathlib, what was new

Mathlib. `field_simp`.

Seabed (reused). The third- and fourth-cumulant theorems¹, moment-normalisation, and the `iteratedDeriv` partition identity — all from this session’s arc.

New. The two moment-form theorems. About 90 lines.

6 Lessons

- **State the headline in the reader’s basis.** The $G_n(0)$ -rational forms are what the proofs produce; the $\langle u^k \rangle_0$ forms are what a physicist reads. A one-helper moment-normalisation rewrite bridges them.
- **Probe whether `field_simp` closes before appending `ring`.** Its terminal behaviour is identity-dependent; a stray `ring` on an already-closed goal is an error, not a no-op.

7 Follow-ups

- **Promote `amgm.t_abs_x` to `lean-common`** — the last open laplace-arc follow-up; load-bearing across the whole arc.
- **Higher cumulants / a fresh seabed.** The anharmonic 1D cumulant programme is now complete through κ_4 in both bases and as $\partial^n \log Z$; further laplace work is marginal, and the next substantial step is a different seabed.

¹`anharmonic_third_cumulant`, `anharmonic_fourth_cumulant`.